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| Chain length(or) no. of carbon atoms | Root word | Chain length(or) no. of carbon atom | Root word |
|---|---|---|---|
| $C_{12}$ | Dodec- | $C_{60}$ | Hexacont- |
| $C_{13}$ | Tridec - | $C_{70}$ | Heptacont- |
| $C_{14}$ | Tetradec- | $C_{80}$ | Octacont- |
| $C_{15}$ | Pentadec- | $C_{90}$ | Nonacont- |
| $C_{16}$ | Hexadec- | $C_{100}$ | Hect- |

**Suffix:** There are two types of suffix. They are primary suffix and secondary suffix

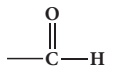
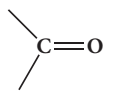
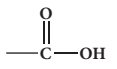
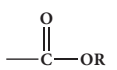
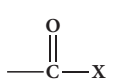
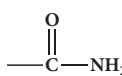
**Primary suffix:** It denotes the saturation/ unsaturation of organic compounds. It is added immediately after the root word. Primary suffix for various saturated and unsaturated carbon chains are as follows:

### Table 11.3 Primary suffix for various saturated and unsaturated carbon chains

| Name & type of carbon chain | Primary suffix |
|---|---|
| Saturated, C-C | ane |
| Unsaturated carbon chain | |
| one C=C bond | ene |
| Two C=C bonds | diene |
| Three C=C bonds | triene |
| One C≡C bond | yne |
| Two C≡C bonds | diyne |

**Secondary suffix:** It is used to denote the nature of functional group present in the organic compound. It is added to the primary suffix by removig its terminal 'e'. Secondary suffix names for some functional groups is listed below in table 11.4

### Table 11.4 Secondary sufix and prefixes for some functional groups:

| Class of organic compounds | Functional group | Prefix | Secondary suffix |
|---|---|---|---|
| Alcohols | —OH | hydroxy- | -ol |
| Thioalcohols | —SH | mercapto | -thiol |
| Aldehydes |  | formyl- | -al |
| Ketones |  | oxo- | -one |
| Carboxylic acid |  | carboxy- | -oic acid |
| Esters |  | Alkoxy–Carbonyl | -oate |
| Acid chlorids |  | chloro-carbonyl | -oyl chloride |
| Acid amines |  | Carbam oyl | -amide |
| Amines | —$NH_2$ | amino- | -amine |
| Nitriles | —C≡N | cyano- | -nitrile |
| Sulphonic acid | —$SO_3H$ | sulpho- | -sulphonic acid |